

Experiment 5

Problem Statement:

Content analysis (text, emoticons, image, audio, video) based social media analytics: Location analysis, Trend analysis, Hashtag popularity analysis, Sentiment analysis for given dataset, User engagement analysis. Analyze content types (text, emoticons, image, audio, video). Use techniques Sentiment analysis with Hugging face Transformers, Trend Analysis via time-series tools (e.g. Prophet) and Hashtag analysis using tools like hashtagify

Theory:

Social media analytics provides valuable insights into user engagement, trends, and audience sentiment. This document explores various analytical techniques such as location analysis, trend analysis, hashtag popularity, sentiment analysis, user engagement, and content-type analysis. By leveraging advanced tools like Hugging Face Transformers, Prophet, and Hashtagify, we extract meaningful patterns from different forms of social media content, including text, emoticons, images, audio, and video.

Location Analysis:

Location analysis helps in understanding the geographical distribution of social media activity. By extracting location metadata from posts and user profiles, businesses can identify engagement hotspots and regional trends. Tools like GeoPandas and Folium enable the visualization of user interactions on a map, offering insights into location-based marketing strategies and consumer preferences.

Trend Analysis:

Trend analysis focuses on identifying how user engagement evolves over time. The dataset is analyzed to detect daily and hourly variations in user activity. By applying **time-series analysis with Prophet**, trends in the volume of comments and posts are visualized. Line plots reveal peak activity periods, helping businesses optimize their content publishing schedule to align with audience engagement patterns.

Hashtag Popularity Analysis:

Hashtags play a critical role in content discoverability and trend formation. The analysis involves extracting hashtags from user comments and computing their frequency using **NLTK and Hashtagify tools**. The most frequently used hashtags are visualized using a horizontal bar chart, highlighting which topics are trending. This insight helps marketers strategically use hashtags to increase post reach and engagement.

Sentiment Analysis:

Sentiment analysis assesses the emotional tone of social media discussions. The **VADER SentimentIntensityAnalyzer from NLTK** is used to classify posts as positive, negative, or neutral. By analyzing sentiment trends over time, businesses can track public perception, detect sentiment shifts, and adjust their engagement strategies to better align with audience sentiments.

User Engagement Analysis:

User engagement analysis evaluates how actively users interact with content through likes, comments, and shares. By computing the number of comments per user and visualizing engagement levels, this analysis identifies highly engaged users and top contributors. A network-based approach using **NetworkX** can further reveal influential users who drive discussions, helping businesses target key audience segments more effectively.

Content-type Analysis:

Different content formats (text, images, audio, and video) have varying impacts on user engagement. The analysis classifies content by detecting keywords related to images (e.g., JPG, PNG), videos (e.g., MP4, YouTube), and audio (e.g., MP3, podcast). Using **matplotlib and pie charts**, the proportion of different content types is visualized. This helps businesses understand which formats perform best and optimize content strategies accordingly.